How Racial Composition Shapes Local Investment in Education*

Voter–Student Mismatch and Texas School Bond Elections, 2014–2025

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Abstract

In school bond elections, adult voters decide whether to tax themselves for facilities that will mainly serve children, yet in many districts the electorate and the student body differ sharply in racial composition. We ask whether that difference helps explain which bonds pass. Building on a model of collective action in which support depends on both group shares and racial fragmentation, we measure the racial distance between the electorate and the student body with the hybrid fragmentation index (HFI) of Kurtz et al. (2024): the probability that a randomly chosen eligible voter and a randomly chosen student belong to different racial or ethnic groups. Using 1,941 Texas school bond propositions from 2014 to 2025, with adults measured by the citizen voting-age population and students by state enrollment records, both annually, we estimate passage models with district and election-year fixed effects. A ten-percentage-point increase in the HFI is associated with a 13.6-percentage-point lower probability that a bond passes. Fragmentation measured within the adult population or within the student body alone is not statistically significant in the same specification. We also find that voters have become less willing over the period to approve new bonds in districts already carrying high debt-service tax rates. The estimates are associations, not causal effects.

Keywords: school bond elections; racial fragmentation; local public goods; education finance; hybrid fragmentation index; Texas.

JEL classification: H41; H75; I22; D72; J15.

*This paper uses public administrative records from the Texas Bond Review Board, the Texas Education Agency, and the U.S. Census Bureau; none of these agencies bears responsibility for the analysis or conclusions. Replication code and the constructed district–year panel are available from the authors. Any remaining errors are our own.

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1 Introduction

School district bonds are the main way U.S. school districts pay for long-lived capital — new buildings, renovations, technology, athletic facilities — and in most states a district cannot borrow without first winning a referendum. Bond elections are therefore one of the few high-stakes public finance decisions made directly by voters, and they separate, by construction, those who pay from those who benefit. The cost falls on local property owners and is spread over two or three decades; the benefit goes to current and future students. In racially diverse places such as Texas, these two groups often look very different: the adults who vote on a bond and the students it would serve can differ sharply in racial and ethnic composition. This paper asks whether the racial distance between the electorate and the student body helps explain which school bonds pass.

Two literatures bear on this question, but neither answers it. The first, in economics, shows that racial and ethnic diversity reduces support for local public goods (Alesina et al., 1999; Alesina and La Ferrara, 2000; Vigdor, 2004; Miguel and Gugerty, 2005). It measures diversity *within* one population, with a fractionalization index, and so cannot capture a gap *between* two populations. The second asks directly whether those who pay for schools resemble those who attend them, but it reduces the comparison to a single dimension: Poterba (1997) uses the difference between the nonwhite shares of the school-age and elderly populations, Brunner and Johnson (2016) interact voter age with the Hispanic share of the college-age population, and Kogan et al. (2021a) compare the White share of school board voters with the White share of students. Studies of school bond elections themselves usually enter race as separate group shares, and where they use a diversity index they compute it for one population only — students in Alvord and Rauscher (2021) and Boroomand and Kaul (2026), all residents in Jones and Zhan (2020). The two studies closest to ours in setting do not measure the gap either: Bowers and Lee (2013) have no measure of adult racial composition, and Berriochoa (2022) enters student and county-population racial ratios as separate variables. To our knowledge, no study of bond elections measures the distance between the full racial distributions of the voters who decide a bond and the students it serves.

We fill this gap with a measure that comes from a model. Following Vigdor (2004), a voter's support for a collective good depends on her own characteristics, on community characteristics, and on the share of her own group in the community. Aggregated to the community, this produces two demographic terms that belong in the same equation: the group shares and a fragmentation term. Kurtz et al. (2024) extend the model to settings in which the people who pay and the people who benefit are different populations, as they are in a bond election. The fragmentation term then becomes the hybrid fragmentation index,

$$HFI = 1 - \sum_{r} \sigma_r^{(c)} \, \sigma_r^{(s)},$$

where $\sigma_r^{(c)}$ and $\sigma_r^{(s)}$ are the shares of group r among eligible adults and among students. The HFI is the probability that a randomly chosen adult and a randomly chosen student belong to different groups. If voters are more willing to pay for a public good when its beneficiaries belong to their own group, as the model assumes, support for a bond should fall as the HFI rises. A within-population index can miss exactly the case that matters. If adults are 70% White and 30% Hispanic, the fractionalization index is 0.42 whether students are 70% White and 30% Hispanic or 30% White and 70% Hispanic; the HFI is 0.42 in the first district and 0.58 in the second.

We apply this framework to Texas independent school district bond propositions from 2014 through 2025, using election records from the Texas Bond Review Board. The estimation sample contains 1,941 propositions in 495 districts. We measure the adult side with the citizen voting-age population (CVAP) — the residents actually eligible to vote — and the student side with Texas Education Agency enrollment, both annually at the district level. Following the model, every specification includes the adult racial shares together with a fragmentation measure, along with bond characteristics, district characteristics, and district and election-year fixed effects. We then compare, in otherwise identical specifications, three fragmentation measures: the HFI, fractionalization of adults, and fractionalization of students.

Our main finding is that districts with a higher HFI are less likely to pass bonds: a ten-percentage-point increase in the HFI is associated with a 13.6-percentage-point lower probability of passage. Fractionalization measured within adults or within students alone is also negatively related to passage but is not statistically significant in the same specification. We also find that voters have become less willing over the period to approve new bonds in districts that already carry high debt-service tax rates. These are within-district associations, not causal effects, and they describe only the elections that districts chose to hold; Sections 7 through 9 report where the HFI result is weaker and what the design cannot rule out.

The paper makes three contributions. First, it connects the economics of diversity and public goods with the literature on who pays for schools and who attends them. To our knowledge, it is the first study of school finance elections to measure the racial distance between voters and students with a cross-population index, and to compare that index with within-population indices in the same specification. Because the specification follows from a model of voter behavior, the comparison tests a prediction rather than a choice among variables. Second, it improves how the two populations are measured. Adults are measured by CVAP, which excludes non-citizens and minors, rather than by all residents, the county population or an age bracket, and both populations are measured every year at the district level. Earlier Texas studies freeze adult demographics at a census (Bowers and Lee, 2013) or measure them at the county level (Berriochoa, 2022). Because actual voters are whiter still than the eligible population (Kogan et al., 2021a), a CVAP-based measure, if anything, understates the distance between those who vote and the students they vote for. Third, it provides

evidence for a recent and eventful period. Existing studies of race and Texas bond elections end by 2018–19; our data run to 2025 and span House Bill 3, the 2019 requirement that bond ballots state the property tax increase (Heller and Lyon, 2026), and the pandemic. Texas also holds the passage threshold at a simple majority throughout, so passage means the same thing in every observation. The finding on debt-service tax burden adds to the literature on age and school spending (Poterba, 1997; Harris et al., 2001; Kogan et al., 2018): in our data, an apparent shift over time in how the elderly share relates to passage is better described as growing resistance to new debt where existing debt is already high. The data favor this reading but cannot rule out the other.

The paper proceeds as follows. Section 2 reviews the related literature. Section 3 presents the model and the indices. Section 4 describes the Texas institutional setting and Section 5 the data. Section 6 sets out the estimating equation, Section 7 presents the results, and Section 8 reports robustness checks. Section 9 discusses interpretation and limitations, and Section 10 concludes.

2 Related Literature

Four literatures meet in this paper: the economics of ethnic diversity and public goods, the political economy of who pays for schools and who attends them, the effect of age structure on school spending, and the empirical study of school bond referenda. We take them in turn and close with a precise statement of what we add and what we do not.

2.1 Diversity and local public goods

The modern literature begins with Alesina et al. (1999), who show across U.S. cities, metropolitan areas and urban counties that the share of spending devoted to productive public goods — education, roads, sewers, trash collection — falls as ethnic fractionalization rises. Their measure, the probability that two randomly drawn residents belong to different groups, is the Herfindahl form we build on, and Alesina et al. (2003) is its standard methodological reference. A large body of work extends the finding to contributions and participation in heterogeneous communities (Alesina and La Ferrara, 2000; Miguel and Gugerty, 2005; Vigdor, 2004; Hungerman, 2009).

The finding does not travel cleanly to school districts. Lee et al. (2016), studying 1,286 school districts and a set of cities over 2000–2010, find that at the school-district level the heterogeneity coefficients are positive: diversity does not reduce per-pupil education spending in their data. Wimmer (2016) argues on cross-national data that the diversity–public goods correlation is confounded by inherited state capacity; his mechanism does not transfer to Texas school districts, but his point that the correlation is confound-prone does. Hopkins (2009), using Massachusetts municipalities (1995–1999) and Texas school districts (1999–2002), finds that diverse places are not in general less willing to raise taxes; what matters is sudden demographic *change*. Texas districts that become more diverse are about 2.1 points less likely to hold a tax-rate rollback election at all, and in Massachusetts

the effect of rapid diversification is concentrated on votes for long-term capital projects.

Three studies relate diversity to school funding elections directly. Jones and Zhan (2020) study California parcel tax elections from 2000 to 2015 with a fractionalization index computed over all district residents: their OLS estimates are negative and significant, but under a shift-share instrument the coefficient falls to less than a quarter of its OLS magnitude and loses significance once income controls are added. Alvord and Rauscher (2021), studying 1,753 California school funding measures in 679 districts from 1995 to 2014, compute a diversity index over *students* only. Their results depend on the measure type and the design. In their within-district models a larger Black student share raises the probability that a bond passes and the diversity index is unrelated to bond passage; in their instrumented models the Black share still raises bond passage and lowers property-tax passage, while student diversity *lowers* bond passage and raises property-tax passage. Boroomand and Kaul (2026) assemble a national district panel for 2009–10 to 2018–19 with district and year fixed effects, a Herfindahl index over six student racial groups, and two outcomes: local revenue per pupil and the voter-approved bond amount per pupil, coded zero where no bond was proposed or where it failed. A one-standard-deviation increase in student diversity is associated with about 1.9% less local revenue and about 7% less approved bond funding per pupil, concentrated in districts with low residential integration or low social connectedness, and the estimates hold under a Bartik-style immigration instrument. Their design — district fixed effects, a fractionalization index, voter-approved bonds — is the closest in structure to ours, and it means that within-district estimates of diversity on bond outcomes are not new. Their index, like Alvord and Rauscher's, is defined over students alone and contains no information about the adults who vote.

There is, in short, no settled sign in this literature for our outcome: it depends on the population over which diversity is measured, on whether the outcome is passage, an approved amount or spending, and on the design. We do not present our own negative coefficient as confirming a consensus.

2.2 Who pays and who benefits

A second strand asks not how diverse a community is but whether the people financing a public good resemble the people receiving it. Poterba (1997) is the origin. Using a panel of U.S. states from 1960 to 1990, he estimates an elasticity of per-child education spending with respect to the elderly share of about -0.25, and — central for us — constructs an explicit two-population racial variable: the nonwhite share of the school-age population minus the nonwhite share of the population aged 65 and over. He reports that this gap averaged 6.5 percentage points, peaked at 23.2 in Florida, and widened by roughly six points between 1960 and 1990. Poterba therefore did not merely observe the phenomenon; he measured it, and the contribution we claim below is stated relative to his variable.

Figlio and Fletcher (2012) take the comparison to school-district outcomes. They find that the share of elderly residents who have aged in place is negatively related to support for public schooling, most strongly in metropolitan districts where the school-age population is more heavily nonwhite relative to the elderly; their racial-mismatch variable is, as summarized by Schmidt (2026), the ratio of the two nonwhite shares. Brunner and Johnson (2016) supply the behavioral counterpart, interacting individual voter age with the Hispanic share of the county's college-age population across roughly 8,200 California survey respondents and precinct-level community college bond returns. Older white voters are approximately 24 percentage points less likely to support higher education funding where 69% of the college-age population is Hispanic; the relationship is absent among older nonwhite voters and fails to appear on twenty-six of twenty-seven unrelated statewide propositions. Brunner and Ross (2010) define a distinct low-demand bloc — voters over 45 without children — whose demand for school spending they estimate at about 45% below that of other voters. Ajilore (2009) measures fractionalization within the elderly population specifically and finds it associated with lower per-pupil spending and local revenue.

Three recent studies measure the voter and student sides directly. Kogan et al. (2021a) use validated turnout from voter files for school board elections in California, Illinois, Ohio and Oklahoma (2008–2016) and show that electorates are far whiter than the students they govern: the typical electorate is majority White once White enrollment reaches only about 20%, and in majority-nonwhite districts the average electorate is 59–90% White. They trace this "representational gap" to both eligibility and turnout and relate it to the White–Hispanic achievement gap, without claiming a causal effect. Their gap is a difference in White shares. Berriochoa (2022) studies Texas school bond elections from 2004 to 2016, using the same Texas Bond Review Board source we use. Her outcome is a district's annual bond passage rate, and racial composition enters as Hispanic-to-White and non-White-to-White ratios, separately for students and for the whole county population, each measured annually. Within counties, a higher Hispanic-to-White ratio in the county population is associated with a lower passage rate — the same sign as our estimate — while the between-county association is positive and the non-White ratio's association reverses between urban and rural counties. Schmidt (2026) enters adult and student racial shares, together with a fractionalization index for each population, in one cross-sectional regression of local revenue per student across 10,537 unified districts in 2018–19, and finds that the two populations carry different associations and that the pattern depends on whether White adults are a large majority.

Two further studies make the same point in our state and in its most extreme form. In the 1996 Houston bond election studied by Tedin et al. (2001), registered voters were about 57% White while students were 11% White; the bond failed, and age reduced support only among White voters. Card et al. (2025) model the limiting case — a virtually all-White electorate in 1940s Mississippi choosing spending for Black and White students — in which the cost to the electorate of a dollar per

Black student rises with the Black share. A recent review identifies the racial difference between the adult or eligible-voter population and the school-age population as a determinant of local revenue that school-finance models typically ignore (Baker and Weathers, 2026).

One feature of the student side deserves note because it bears on what any such comparison measures. Enrolled students are not the same population as resident children: Bischoff and Tach (2018) show that the White share of enrolled pupils tracks the White share of resident children roughly one for one on average, but much less closely in majority-Black attendance zones, and less closely where income inequality and private and magnet school options are greater. Part of any measured gap between adults and enrolled students therefore reflects which families use the public schools.

The common feature of all of these measures is that the comparison is reduced to separate shares or to a single dimension — a difference or ratio of nonwhite shares, one group's share interacted with age, a difference in White shares, ratios to White entered separately, or group shares entered side by side. None compares the full racial distributions of the two populations in a single measure, which is what the index in Section 3, introduced by Kurtz et al. (2024), does.

2.3 Age structure and school spending

Because our specification includes the elderly share and because that coefficient behaves in an initially surprising way, the relevant background deserves a note. Harris et al. (2001) estimate an elasticity of district revenue per pupil with respect to the 65-and-over share of -0.096 using district fixed effects and year-specific state effects — about 40% of Poterba's state-level estimate. The gap is attributed to the elderly being far less willing to fund state than local spending, and Brunner and Balsdon (2004) confirm the same asymmetry directly in survey data: support for school spending declines with age, but older voters are more willing to back local school spending than state spending. Ladd and Murray (2001) find no distinguishable within-county elderly effect at all. Berkman and Plutzer (2004), using more than 9,000 districts, show the elderly are heterogeneous: longstanding older residents support school spending while elderly in-migrants depress it. A small, imprecisely estimated district-level elderly coefficient is therefore the documented expectation in our setting, not an anomaly. Cattaneo and Wolter (2009) and Grob and Wolter (2007) provide the European survey evidence.

Texas adds a specific twist. Kogan et al. (2018), studying 10,353 school tax and bond referenda in four states including Texas, show that election timing changes who votes mainly through age: in Texas special elections the 65-and-over share of the electorate is about 16 points higher. Yet in Texas, unlike the other three states, passage is *higher* in special elections, which the authors attribute to the state's school tax freeze for homeowners aged 65 and over. An older electorate need not be a less supportive one where the elderly are insulated from the tax cost.

2.4 School bond elections

Bond elections matter for racial and spatial funding gaps because capital is financed differently from operations. Most state formulas equalize operating budgets but leave facilities to separate state programs or local bonds, so capital outlays remain regressive and dependent on local approval (Shores and Lee, 2026). State finance reforms since 1990 have narrowed funding gaps by district income but not by race or ethnicity (Rauscher and Fiel, 2026), which leaves local decisions such as bond referenda as a place where demographic politics continue to bite.

Bowers and Lee (2013) study 2,224 Texas bond propositions from 1997 to 2009, drawn from the same Texas Bond Review Board source we use, with the bond attempt ("float") as the unit of analysis and a 77% passage rate. They estimate a logistic discrete-time hazard with pseudo-intercepts for first, second and third floats and with calendar-year dummies, and find the 65-and-over share negatively related to passage (-0.044, odds ratio 0.957) and Asian and Hispanic student shares positively related. They include no district fixed effects and report no clustering, and their community demographics — the 65-and-over share and the college-educated share — are taken from the 2000 Census and held fixed across the entire thirteen-year panel, a limitation they name themselves. They have no adult racial-composition variable. Berriochoa (2022), discussed above, is the other study relating racial composition to Texas bond passage; her adult side is measured annually but for the county population as a whole.

On information and salience, Brunner et al. (2018) compare Minnesota and Wisconsin bond referenda from 2008 to 2017, exploiting Minnesota's requirement that the ballot state the property tax consequence in bold capitals, and find the yes-share falls by roughly 4 to 7 points and passage by about 10; race enters only as a fraction-nonwhite control. The survey-experimental companion (Brunner et al., 2021) reproduces the reduction under every tested wording. Heller and Lyon (2026) study Texas's own 2019 requirement that school bond ballots state "THIS IS A PROPERTY TAX INCREASE." Using municipal bonds as a control group across 5,053 Texas bond elections from 2012 to 2024, also drawn from the Bond Review Board, they estimate that the disclaimer reduced school bond passage by about nine percentage points. Their design covers our state, our period and our data source; identifying the disclosure effect is not our aim, but election-year fixed effects absorb it only to the extent that it is common across districts.

A separate literature treats bond passage as a treatment rather than an outcome (Cellini et al., 2010; Martorell et al., 2016; Baron, 2022; Abott et al., 2020; Biasi et al., 2025; Rauscher, 2020; Rauscher et al., 2025). Several include Texas: Martorell et al. (2016) study nearly 1,400 capital campaigns across 748 Texas districts and justify the single-state focus on the grounds that Texas is large and diverse, Abott et al. (2020) include Texas among seven states, and Rauscher et al. (2025) include Texas bond elections through 2018. In all of them racial composition is, at most, a control or a dimension of heterogeneity in the effect of passage, not a determinant of it.

On electoral institutions, Romer et al. (1992) establish the importance of the reversion level facing voters when a proposal fails; Anzia (2011) shows that off-cycle timing advantages organized interests, with Texas-specific evidence in Anzia (2012); and Grosz and Milton (2025) exploit California's Proposition 39 threshold change. Bechard et al. (2024) find in a survey experiment that additional bonds on the ballot and a lower ballot position both reduce support — direct motivation for two of our controls. On the supply side, Balsdon et al. (2003) model the proposal decision jointly with proposed spending and the vote and find that school boards behave like risk-averse, budget-maximizing agenda setters, and Kogan et al. (2021b) show in California that the election of a minority board member is followed by larger bond proposals. Together with Hopkins (2009), these studies establish that which bonds reach the ballot is itself related to a district's composition and governance.

Finally, the later part of our window coincides with pandemic-era federal aid. Clemens et al. (2026) find that districts just above the Title I threshold that determined ESSER allocations, which received about $388 more ESSER funding per pupil, reduced local revenue by about $907 per pupil over three years, roughly 80% of it through property taxes, although their identifying sample is small, low-poverty districts largely outside the South. Because ESSER was allocated by poverty, which is correlated with student composition, this is relevant to the period pattern we report in Section 8.3.

2.5 What this paper adds

Stated precisely, our contribution is one of measurement, population and setting rather than of identification or estimator.

On *measurement*, prior work reduces the voter–student comparison to one dimension or to separate shares: Poterba uses a difference in nonwhite shares, Figlio and Fletcher a ratio of nonwhite shares, Brunner and Johnson an interaction of age with a single group's share, Kogan et al. (2021a) a difference in White shares, Berriochoa ratios to White entered separately for each population, and Schmidt group shares entered side by side. Alvord and Rauscher and Boroomand and Kaul compute an index over students only. We use an index defined over the full racial distributions of both populations simultaneously, which registers divergence that a single-dimension measure cannot — the leading case being two populations with identical nonwhite shares whose minority compositions do not overlap. We should be equally precise about two limits. First, an index of total distance between distributions cannot say *which* pairing drives it: Brunner and Johnson can state that older white voters withdraw support for community-college bonds where the college-age population is more Hispanic, and we cannot. Second, the index can be high either because each population is diverse or because the two are composed differently, and our data cannot cleanly separate the two (Section 9.1). The measurement contribution is therefore an index that summarizes

both populations jointly; whether mismatch as such moves voters is a question our data cannot settle.

On *population*, we measure the adult side with the citizen voting-age population. Poterba uses residents aged 65 and over; Bowers and Lee use a 2000 Census age share; Jones and Zhan use all residents; Berriochoa uses the whole county population and Schmidt all adults; Alvord and Rauscher and Boroomand and Kaul have no adult racial measure at all. CVAP excludes non-citizens and minors, precisely the residents who appear in a residence-based measure but cannot vote, and in Texas districts with substantial non-citizen populations the two measures diverge materially. CVAP is not the electorate itself: Kogan et al. (2021a) show that turnout widens the racial gap between voters and students beyond what eligibility alone implies, so a CVAP-based measure plausibly understates the gap among those who actually vote. Boroomand and Kaul (2026) also caution against five-year ACS estimates for district-year measures; we take up the smoothing this implies in Section 5.2.

On *frequency*, both populations are measured annually at the district level. Bowers and Lee's adult data cannot move at all; Jones and Zhan state explicitly that their year-to-year variation comes from linear interpolation between censuses. Berriochoa also measures both sides annually, but her adult side is the county population rather than the district electorate. Annual measurement on both sides is what makes the time-trend interactions in Section 7 estimable. We note the obvious rejoinder in Section 5: CVAP arrives in overlapping five-year ACS vintages, so our own "annual" variation is smoothed, if less severely.

On *setting and period*, Texas school bonds face a simple-majority threshold throughout, so passage means the same thing in every observation — unlike Alvord and Rauscher, whose threshold varies by measure type and changed mid-sample. Our 2014–2025 window overlaps the end of the existing evidence: Berriochoa's Texas data end in 2016, the Texas referenda in Kogan et al. (2018) around 2015, Rauscher et al.'s Texas bond elections in 2018, and Boroomand and Kaul's national panel in 2018–19. Heller and Lyon (2026) cover our years but study the disclosure rule rather than composition. To our knowledge, no study relates district racial composition to bond outcomes after 2018–19, and the later part of our window spans House Bill 3, the 2019 ballot-disclosure requirement, rate compression and the pandemic.

What we do not add is identification. Alvord and Rauscher (2021), Jones and Zhan (2020) and Boroomand and Kaul (2026) instrument for composition, Brunner and Johnson (2016) run a falsification test across unrelated propositions, and Heller and Lyon (2026) have a control group; we have none of these. Our econometric frame — district and year fixed effects, district-clustered errors, a linear probability model with logit as a check — closely resembles the one Alvord and Rauscher (2021) already use, including their justification for preferring the linear model, and shares its fixed-effect structure with Boroomand and Kaul (2026). We think it is better to say so than to

imply methodological novelty that a referee would correct.

3 Conceptual Framework

This section does two things. It derives, from a standard model of collective action, why a fragmentation term belongs in an estimating equation for bond approval alongside racial shares. And it extends that model to a setting with two demographically distinct populations and obtains the hybrid fragmentation index.

3.1 From individual behavior to a fragmentation term

Vigdor (2004) models an individual's participation in a collective action as a function of individual attributes, community characteristics, and the individual's own group share in the local population. For individual i of racial group r in community c,

$$\Pr(Y_{irc} = 1) = \alpha_r + \beta X_{irc} + \gamma Z_c + \delta_r \sigma_{rc}, \tag{3.1}$$

where X_{irc} denotes individual attributes, Z_c community characteristics, and σ_{rc} the population share of group r in community c. The parameter δ_r captures within-group affinity: if $\delta_r > 0$, an individual is more willing to support a collective action when a larger share of the beneficiaries belongs to their own group. The subscript on δ_r allows the strength of that affinity to differ across groups, while β and γ are common.

Aggregating (3.1) to the community level by summing over groups weighted by their shares, with a common affinity parameter $\delta_r = \delta$, yields

$$\overline{\Pr}(Y = 1)_c = \sum_r \alpha_r \sigma_{rc} + \beta \bar{X}_c + \gamma Z_c + \delta \sum_r \sigma_{rc}^2, \tag{3.2}$$

where $\bar{X}_c$ is the community mean of individual characteristics. The final term, $\sum_r \sigma_{rc}^2$, is the Herfindahl concentration index, and it is a linear transformation of the fractionalization index $FI_c = 1 - \sum_r \sigma_{rc}^2$.

Equation (3.2) delivers the first requirement for a valid estimating equation, and it is a requirement about specification rather than about which variable is interesting. Aggregating individual behavior generates a fragmentation term *mechanically*, alongside the racial shares. An equation that includes shares but omits fragmentation, or fragmentation but omits shares, is misspecified relative to the aggregated model. This is why every specification we report includes the adult racial shares and the index together, and why the practice in much of the bond literature of entering shares alone is not innocuous.

3.2 Two populations: donors and recipients

Equation (3.2) describes a single population that both acts and benefits. School bonds do not fit that description: adults vote and pay, students benefit. Kurtz et al. (2024) extend the logic to exactly this configuration, in their case the adoption of weekend ("BackPack") food assistance programs, where adult donors and child recipients differ demographically. Writing $\sigma_r^{(c)}$ for the share of group r among community adults and $\sigma_r^{(s)}$ for its share among the beneficiary population, their aggregated outcome equation is

$$\overline{\Pr}(Y=1)_{cs} = \beta_1 S_s + \beta_2 C_c + \gamma \bar{X}_c + \sum_r \theta_r \sigma_r^{(c)} + \delta \sum_r \sigma_r^{(c)} \sigma_r^{(s)}, \tag{3.3}$$

where S_s and C_c denote beneficiary-side and community-side characteristics. Two features matter. The cross-population term $\sum_r \sigma_r^{(c)} \sigma_r^{(s)}$ replaces the within-population Herfindahl of (3.2): it is the probability that a randomly drawn adult and a randomly drawn student belong to the *same* group. And the racial shares that must be controlled for are the *community* shares $\sigma_r^{(c)}$ — the group shares of the population that acts. Our specification follows (3.3) directly: adult racial shares enter as controls, and the cross-population term enters as the variable of interest.

Defining the hybrid fragmentation index as one minus that term,

$$HFI = 1 - \sum_r \sigma_r^{(c)} \sigma_r^{(s)}, \tag{3.4}$$

gives a quantity that is increasing in demographic distance between the two populations. It equals zero only when both the entire student body and the entire adult population belong to the same single group, and approaches one when students belong entirely to groups to which adults do not belong. In the words of Kurtz et al. (2024), the index "declines as community members shift from a group that is less prevalent in the school to one that is more prevalent" — that is, it falls as who pays comes to resemble who benefits. The standard fractionalization index is the special case of (3.4) in which the two populations coincide.

3.3 What the index registers that a within-population index cannot

Table 1 makes the distinction concrete. Each case is constructed so that the adult and student fractionalization indices are equal to one another, isolating the effect of *alignment* from the effect of *diversity*.

Cases A through D use two groups and vary the degree of reversal between the distributions. In Case A the populations are identical and HFI equals the common FI of 0.42. In Case B the compositions are exactly reversed: both fractionalization indices are unchanged at 0.42, while HFI rises to 0.58. In Case C, the extreme, the two populations share no group at all: both FI values

are zero — each population is perfectly homogeneous — while HFI reaches its maximum of one. This is the clearest demonstration that FI and HFI are not measuring the same object: the case in which the two fractionalization indices report the *least* possible heterogeneity is precisely the case of maximal mismatch.

Case E is the configuration that motivates the paper and that a single-dimension difference cannot detect. The adult population is 70% White and 30% Black; the student population is 70% White and 30% Hispanic. The nonwhite share is 30% in both populations, so the Poterba (1997) difference is exactly zero, and both fractionalization indices equal 0.42. Yet the minority populations do not overlap at all, and HFI registers this at 0.51.

Case F relaxes the equal-FI construction to show that HFI is not simply the larger of the two indices. Here adults are evenly split while students are highly concentrated, so $FI^c = 0.50$ and $FI^s = 0.18$. The index takes the value 0.50, coinciding numerically with the adult index, even though a substantial mismatch is present. The reason is that when adults are split evenly between two groups, a randomly chosen adult matches a randomly chosen student with probability one-half whatever the student composition, so the index equals 0.50 however the students are distributed between those two groups.

Table 1: Numerical Illustration: Fractionalization and Hybrid Fragmentation Under Alternative Adult–Student Compositions

Case	Adult shares	Student shares	FI^c	FI^s	HFI
Panel A: Two groups, equal fractionalization by construction					
A	70% White, 30% Hispanic	70% White, 30% Hispanic	0.42	0.42	0.42
B	70% White, 30% Hispanic	30% White, 70% Hispanic	0.42	0.42	0.58
C	100% White	100% Hispanic	0.00	0.00	1.00
D	80% White, 20% Hispanic	20% White, 80% Hispanic	0.32	0.32	0.68
Panel B: Non-overlapping minorities and unequal fractionalization					
E	70% White, 30% Black	70% White, 30% Hispanic	0.42	0.42	0.51
F	50% White, 50% Hispanic	90% White, 10% Hispanic	0.50	0.18	0.50

Notes: $FI^c = 1-\sum_r (\sigma_r^{(c)})^2$ and $FI^s = 1-\sum_r (\sigma_r^{(s)})^2$ are the within-population fractionalization indices for adults and students; $HFI = 1-\sum_r \sigma_r^{(c)} \sigma_r^{(s)}$ is the hybrid fragmentation index of Kurtz et al. (2024). In Case E the nonwhite share is 30% in both populations, so the Poterba (1997) nonwhite-share difference is exactly zero while HFI registers 0.51. Values are rounded to two decimals.

A related measure deserves mention because it is sometimes proposed in its place. Theil's H, used in this literature by Kitchens (2021), captures segregation — how unevenly racial groups are distributed across subunits such as schools or census tracts, relative to the composition of the whole. When no subunit variation is available, as in district-level election data, Theil's H is mechanically zero. It answers a different question, about distribution across units within one population, and cannot register the representation gap between two populations at the level at which bond decisions are made. A referee may also reasonably ask whether polarization rather than fractionalization is the right functional form, following Montalvo and Reynal-Querol (2005); we return to this in

Section 9.

4 Institutional Setting

Texas offers an unusually clean environment for this question, and three features of its institutions shape the design.

The same passing rule for every bond. Texas Education Code § 45.003(a) requires a simple majority of the votes cast to approve a school district bond, with no supermajority requirement, and this rule applies to every district and every year in our sample. Our outcome is whether a bond passes, so this matters: a bond with 52% support passes everywhere in our data, and "passing" means the same thing for every proposition. That is not true in every setting studied in this literature. Alvord and Rauscher (2021) note that the passing threshold in their California sample varies by measure type and changed during their sample period, and Jones and Zhan (2020) study parcel taxes that require a two-thirds majority.[1]

Who may hold an election. Only traditional independent and consolidated school districts may call bond elections in Texas. Open-enrollment charter schools are independent local education agencies with no bond authority of their own, and we confirm empirically that no charter district appears anywhere in the Texas Bond Review Board election file. A separate category, campus charters under Texas Education Code Chapter 12, Subchapter C — in which a district charters one of its own campuses, or contracts an outside operator to run one under the 2017 SB 1882 partnership program — remains legally part of the district, is counted in district enrollment, and does not affect bond authority. In the Texas Education Agency's fall 2025 school directory there are 104 campus-charter campuses in 21 districts, against 1,057 campuses in 195 open-enrollment and university charter districts. Both types cluster in the largest metropolitan counties: Harris, Dallas, Bexar, Tarrant and Travis counties contain 56% of campus charters and 59% of charter-district campuses, against 35% of all campuses. We treat campus-charter presence as a control and charter proximity as a robustness dimension in Section 8.4.

A tax ceiling for elderly homeowners. Texas freezes the total school district tax bill — including the debt-service portion that bonds fund — for any homeowner aged 65 or over or with a disability (Texas Constitution art. VIII § 1-b(d); Texas Tax Code § 11.26). Once frozen, the bill cannot rise above its first qualifying-year level no matter how many bonds subsequently pass, absent a major improvement to the property. The ceiling is not means-tested and is unaffected by ordinary appreciation in home values. Changes since 2007 have all moved in the same direction: the 2007

[1] House Bill 68, filed in the first called session of the 89th Texas Legislature in 2025, proposed raising the threshold to three-fifths, contingent on a constitutional amendment requiring separate voter approval. It could not have taken effect before 1 January 2026, after our sample period.

amendment authorizing pass-through of rate cuts, the 2015 and 2022 increases in the homestead exemption, and the 2022 and 2023 measures extending House Bill 3's rate compression to already-frozen homeowners. An elderly Texas homeowner's financial exposure to a new bond has therefore weakened monotonically over time.

This institutional detail bears directly on a coefficient. The standard "gray peril" expectation is that a larger elderly share depresses support for school spending, and Bowers and Lee (2013) indeed find a negative elderly coefficient in Texas over 1997–2009. Our pooled estimate is positive and insignificant. The tax ceiling is a plausible reason why elderly Texans might be less oppositional than elderly voters elsewhere, but it cannot by itself explain a *change* within Texas, since the ceiling was already in force throughout Bowers and Lee's window. We take up the time pattern in Section 7.3 and are careful not to over-claim a mechanism we have not tested.

A mid-sample disclosure rule. In 2019, House Bill 3 required every school district bond proposition to carry the statement "THIS IS A PROPERTY TAX INCREASE" (Texas Education Code § 45.003(b-1)), and Senate Bill 30 required separate propositions for certain facilities — among them stadiums seating more than 1,000 spectators, natatoriums, performing arts facilities, teacher housing and technology equipment (§ 45.003(g)). Heller and Lyon (2026) estimate, using non-school municipal bonds as a control group, that the disclaimer reduced school bond passage by roughly nine percentage points. Our election-year fixed effects absorb the average effect of this change, and identifying it is not our aim; we note it because it falls inside our window, because it is one reason passage rates decline over our sample, and because it is a candidate explanation for time-varying patterns we document but cannot separately identify.

5 Data

The analysis rests on a district–year panel that merges bond election records with annually measured adult and student racial composition and a set of district and election controls. This section documents each source, the geographic and temporal matching required to combine them, the construction of the estimation sample, and the resulting descriptive statistics. Appendix A reports the merge diagnostics and match rates in full.

5.1 Bond elections

Election records come from the Texas Bond Review Board's Local Debt Bond Election Results file, restricted to issuers the file classifies as independent school districts (a category that includes consolidated independent school districts), and to propositions that either carried or were defeated. For each proposition the file records the issuer and county, the election date, the amount requested, the stated purpose, and the outcome. The outcome defines our dependent variable, *pass*, equal to

one if the proposition carried. The underlying file covers 1958 onward; the usable window is set by the covariate data described below.

The unit of observation is the *proposition*, not the district-year. Texas districts frequently place several propositions on the same ballot — in our estimation sample, the 1,941 propositions are spread across 1,219 distinct district–election dates, and 434 of those dates — roughly a third — carry more than one proposition. We use the proposition as the unit because that is the level at which passage is determined and at which purpose and ballot position vary, and because it is the unit Bowers and Lee (2013) use in the same setting. It does mean that propositions sharing a ballot face the same electorate on the same day, which we address through district-clustered standard errors.

From the election file we also construct four ballot-mechanics controls, all standard in this literature: an indicator for a May (off-cycle) rather than November election, the number of propositions on that ballot, an indicator for first position on the ballot, and an attempt number. The last is a heuristic rather than a linked bond identifier: it counts election-level attempts, incrementing only when the immediately preceding election in the same district was fully defeated and occurred within 24 months. Districts that pass on the first try, or that wait more than two years, are always coded as first attempts, so the variable is a lower bound on true resubmissions.

5.2 Adult composition: the citizen voting-age population

The adult side is measured from the Census Bureau's Citizen Voting Age Population Special Tabulation, which reports CVAP counts by race and ethnicity at tract level. We use fourteen overlapping five-year ACS vintages, from 2007–2011 through 2020–2024, giving effectively one observation per calendar year from 2011 through 2024. Tracts are allocated to school districts using the Missouri Census Data Center's population-weighted *geocorr* crosswalks, with the 2018 vintage applied through 2019 and the 2022 vintage from 2020 onward — a single switch that coincides with the Census Bureau's 2020 tract boundary redraw, visible in the data as a jump in tract count from 5,265 to 6,896 exactly between the 2015–2019 and 2016–2020 vintages. Between 99.96% and 100% of tract-level CVAP totals are preserved through allocation in every year.

Using CVAP rather than total population or an age bracket is a deliberate choice. CVAP excludes non-citizens and minors — exactly the residents counted by a residence-based measure who cannot cast a ballot — and in Texas districts with substantial non-citizen populations the two measures differ materially. The honest qualification is that CVAP arrives in overlapping five-year windows, so consecutive annual observations share four years of underlying survey data and our "annual" variation is smoothed. This is a milder version of the objection we raise against interpolation between decennial censuses, and we state it in those terms rather than claiming measurement our data does not support.

5.3 Student composition

Student racial composition and total enrollment come from the Texas Education Agency's annual enrollment-by-race files for school years 2011–12 through 2025–26, restricted to traditional ISD and CSD districts. Because open-enrollment charter students are enrolled at their own local education agency in state records, they never entered ISD enrollment counts and require no exclusion. Year-over-year enrollment growth is computed within each district's own chronological sequence; it is necessarily missing in a district's first panel year, which is the single largest source of sample attrition below. The economically disadvantaged share comes from the corresponding TEA annual files; where TEA suppresses a district's count to protect student privacy, recorded in those files as -999, we treat the share as missing.

Both populations are collapsed to four categories: White, Black, Hispanic, and Other non-Hispanic, the last combining American Indian and Alaska Native, Asian, Native Hawaiian and Pacific Islander, and two-or-more-races. We considered separating Asian students and adults into a fifth category and rejected it on empirical grounds: 43% of the propositions in our bond panel come from district-years with zero Asian students, and between 31% and 45% of matched rows have fewer than thirty Asian adults in CVAP. A regressor that is identically zero in more than two-fifths of the sample cannot support district and year fixed effects, so Asian counts remain folded into Other on both sides. Appendix A reports the underlying distributions.

5.4 Timing and district controls

Every covariate is attached with a one-year lag: an election held in year E receives covariates from year $E-1$, the most recent full data a voter could have observed before casting a ballot. A flag in the panel records whether the match is a genuine one-year lag or was clipped because the election predates the start of the covariate data in 2012; only genuine-lag rows are used, which is what restricts the sample to elections from 2013 onward.

Two controls warrant specific comment. The first is *is_tax_rate*. It is not an indicator for a tax-rate election; it is the district's interest-and-sinking (I&S) rate — the continuous debt-service property tax rate in dollars per $100 of assessed valuation — taken from the TEA's School District Adopted Tax Rates file and joined on TEA's exact county-district number rather than by name. Only the I&S rate is kept; the maintenance-and-operations rate is dropped to avoid collinearity. The variable therefore measures the debt-service tax burden a district already carries at the time of the election, and it ranges from zero to roughly $0.50 in our sample.

The second is *pct_65plus* and the remaining ACS controls — median household income, poverty, education, homeownership and age structure — drawn from five-year ACS estimates at tract level via `tidycensus`, restricted to the population aged 18 and over where the underlying table permits (B17001 for poverty, B15001 for education), and crosswalked to districts using the

same population-weighted procedure as CVAP. We began with fourteen controls and trimmed to ten after a collinearity check: the bachelor's-degree share was 0.80 correlated with median income, the poverty rate -0.66 correlated with it, the homeownership share never significant and unstable in sign, and the under-18 share -0.59 correlated with the 65-and-over share. We retain both the TEA economically disadvantaged share and ACS median household income despite their -0.65 correlation, since the first is the field-standard school finance measure and the second the standard ability-to-pay proxy in local public finance.

5.5 Sample construction

Table 2 traces the path from the full election file to the estimation sample. The starting file contains 4,868 propositions that either carried or were defeated; 59 cancelled propositions and 8 with no recorded result are excluded. Three restrictions do the work. Requiring a genuine one-year covariate lag removes pre-2013 elections, cutting the file to 2,519 propositions, of which 19 cannot be assigned a TEA district number and are dropped, leaving 2,500. Requiring at least two propositions per district removes single-observation districts whose fixed effect cannot be identified, leaving 2,232. Complete-case requirements on the right-hand side remove a further 251, of which the great majority — 186 rows — are lost to the first-year enrollment-growth calculation, a mechanical consequence of differencing rather than a data quality problem. Dropping fixed-effect singletons leaves an estimation sample of 1,941 propositions across 495 districts and 1,219 district–election dates, spanning 2014 through 2025.

One point deserves transparency. The restriction to districts with at least two propositions is implemented on propositions, not on distinct election dates, so a district that placed two propositions on a single ballot satisfies it; for such districts the fixed effect is identified only from propositions facing the same electorate on the same day. We flag this as a design choice a referee may reasonably question and report in Section 8 what happens under the stricter date-based rule.

5.6 Descriptive statistics

Table 3 reports the estimation sample. Two-thirds of propositions pass (65.6%), below the 77% Bowers and Lee (2013) report for 1997–2009 and consistent with the decline in passage over our window visible in Figure 2. The median bond requests $22.5 million; the mean, $99.5 million, reflects a small number of very large metropolitan issues.

The demographic rows are the substantive ones. The two populations differ systematically and in the direction the framework anticipates. Adults are 64.2% White and 24.0% Hispanic; students in the same districts are 45.1% White and 41.5% Hispanic. The White share is nineteen points higher among the electorate than among the students it votes for, and the Hispanic share seventeen and a half points lower. This gap is the raw material the index summarizes, and it is large enough that a design measuring only one population would miss most of it.

Table 2: Sample Construction

Restriction	Propositions	Districts	Rows lost at this step
All ISD bond propositions, TBRB file	4,868	947	—
Panel A: Timing and identification restrictions			
Genuine one-year covariate lag (2013 onward)	2,519	823	2,349
Assigned a TEA district number	2,500	823	19
Districts with at least two propositions	2,232	555	268
Panel B: Complete-case requirements, cumulative			
+ HFI and adult racial shares	2,228	555	4
+ bond amount, timing, ballot position, attempt	2,228	555	0
+ enrollment growth, one year	2,042	542	186
+ economically disadvantaged share	1,984	537	58
+ I&S tax rate	1,983	536	1
+ median household income (ACS)	1,983	536	0
+ population 65 and over, bond purpose	1,983	536	0
+ campus-charter control	1,981	535	2
Panel C: Estimation sample			
Drop fixed-effect singletons	**1,941**	**495**	40

Notes: The first row counts propositions by independent school districts in the Texas Bond Review Board file that carried or were defeated; the file's 59 cancelled propositions and 8 with no recorded result are excluded. The estimation sample spans election years 2014 through 2025 and comprises 1,941 bond propositions across 495 districts and 1,219 distinct district–election dates. Complete-case requirements in Panel B are cumulative and are listed in the order the variables enter the estimating equation, so the "rows lost" column reflects each variable's incremental rather than independent cost. Enrollment growth is the largest single cut because a district's first year in the panel has no prior year from which to compute a change. Fixed-effect singletons are districts or election years left with a single observation after complete-case selection, which contribute no within variation and are dropped iteratively.

Consistent with that, the mean hybrid fragmentation index is 0.501, against 0.425 for adult fractionalization and 0.483 for student fractionalization. The three are correlated but far from interchangeable: HFI correlates 0.87 with FI^c and 0.83 with FI^s, while the two within-population indices correlate only 0.56 with each other. Figure 1 shows that all three indices rise over the sample period and that the gap between HFI and the two within-population measures is stable rather than widening.

6 Empirical Strategy

6.1 Estimating equation

Our estimating equation follows directly from equation (3.3). For bond proposition b in district d voted on in year t,

$$\begin{aligned} pass_{bdt} = \beta\, HFI_{d,t-1} + \sum_r \theta_r\, \sigma^{(c)}_{r,d,t-1} + \gamma' X_{bdt} + \lambda' Z_{d,t-1} \\ + \pi_1 \left(age65_{d,t-1} \times \tau_t \right) + \pi_2 \left(taxrate_{d,t-1} \times \tau_t \right) + \delta_d + \eta_t + \varepsilon_{bdt}, \end{aligned} \tag{6.1}$$

Table 3: Descriptive Statistics, Estimation Sample

Variable	Mean	Std. dev.	Min	Max
Panel A: Outcome and bond characteristics				
Bond passes (= 1)	0.656	0.475	0	1
Yes-vote share	0.558	0.137	0.000	1.000
Bond amount ($ millions)	99.5	242.1	0.1	3,960.0
Log bond amount per CVAP	7.629	1.542	1.527	13.167
May (off-cycle) election (= 1)	0.589	0.492	0	1
First proposition on ballot (= 1)	0.628	0.483	0	1
Propositions on the ballot	2.160	1.245	1	6
Attempt number	1.209	0.509	1	5
Panel B: Fragmentation indices				
Hybrid fragmentation index, HFI	0.501	0.162	0.021	0.771
Adult fractionalization, FI^c	0.425	0.158	0.030	0.742
Student fractionalization, FI^s	0.483	0.159	0.000	0.744
Panel C: Adult composition (citizen voting-age population)				
White share	0.642	0.215	0.010	0.963
Black share	0.080	0.091	0.000	0.714
Hispanic share	0.240	0.204	0.002	0.985
Other share	0.038	0.038	0.000	0.282
Panel D: Student composition (TEA enrollment)				
White share	0.451	0.248	0.000	0.950
Black share	0.079	0.100	0.000	0.711
Hispanic share	0.415	0.241	0.022	1.000
Other share	0.055	0.057	0.000	0.578
Panel E: District controls				
Population aged 65 and over	0.157	0.056	0.043	0.401
Economically disadvantaged students	0.539	0.203	0.000	0.957
Enrollment growth, one year	0.015	0.104	−0.199	2.859
I&S tax rate ($ per $100 valuation)	0.254	0.140	0.000	0.500
Median household income ($)	71,523	25,765	23,471	234,414

Notes: $N = 1{,}941$ bond propositions across 495 districts and 1,219 district–election dates, election years 2014–2025. Shares are proportions. All covariates are measured with a one-year lag relative to the election. Bond purpose, not shown, is 51.6% Building, 45.5% Other (which includes refunding), 2.4% Renovations and 0.5% Programs. The three fragmentation indices correlate as follows: $\text{corr}(HFI, FI^c) = 0.87$, $\text{corr}(HFI, FI^s) = 0.83$, $\text{corr}(FI^c, FI^s) = 0.56$.

where $HFI_{d,t-1}$ is the hybrid fragmentation index, $\sigma^{(c)}_{r,d,t-1}$ are the adult CVAP shares of White, Black and Hispanic residents (Other is the omitted reference), X_{bdt} collects proposition-level characteristics — log bond amount per CVAP, May election, number of propositions, first ballot position, attempt number, and purpose fixed effects with Other as the reference — and $Z_{d,t-1}$ collects district covariates: enrollment growth, the economically disadvantaged share, median household income, the elderly share, the I&S tax rate, and a campus-charter indicator. The terms δ_d and η_t are district and election-year fixed effects, τ_t is a linear time trend, and standard errors are clustered by district throughout.

Three features of (6.1) follow from Section 3 rather than from convenience. The adult racial

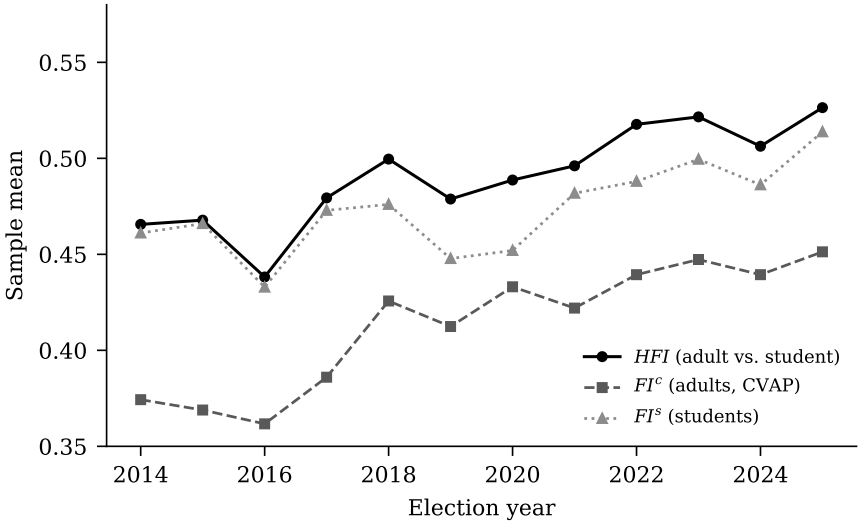


Figure 1: Fragmentation indices over the sample period

Notes: Sample means by election year across the 1,941 propositions in the estimation sample. All three measures rise over the period, reflecting the continued diversification of both the Texas electorate and its student body. The vertical distance between HFI and the within-population indices is broadly stable, indicating that the two populations have diversified together rather than apart.

shares enter alongside the index because the aggregated model generates both, and omitting either misspecifies the equation. The shares are the *adult* shares, as equation (3.3) requires, not the student shares. And the index is not exactly collinear with its own component shares — it is a quadratic rather than linear function of them — so the specification estimates, though with some inflation of the index standard error. We checked that inflation empirically and found it mild: the index standard error is 0.60 with all three shares included against 0.56–0.59 in single-share variants.

6.2 Estimator

We report three estimators for every specification. The linear probability model on the full estimation sample is our headline: its coefficients are directly interpretable in probability points, it accommodates two-way fixed effects without incidental-parameter concerns, and it retains every observation. Alvord and Rauscher (2021) make the same choice on the same grounds in the closest comparable study, citing Ai and Norton (2003) and Mood (2010).

We also report a fixed-effects logit and, critically, a linear model re-estimated on *exactly* the rows the logit retains. The distinction matters here more than it usually does. Conditional logit with district and year fixed effects must drop any district whose outcome never varies — every proposition passed, or every one failed — because such a district contributes no likelihood. In our data that removes a substantial share of observations, taking the logit sample from 1,941 to 1,209. Comparing the full linear model against the logit therefore confounds functional form with sample composition. The third column separates them: any difference between columns (1) and (3) is composition, and any difference between (2) and (3) is functional form.

This matters because the comparison is not flattering. As Section 7.1 reports, the index coefficient grows in magnitude but loses conventional significance when restricted to the logit-retained rows. The full-sample result is partly carried by districts that never vary in passage, which contribute to the within-district demeaning without containing any passage variation of their own. We report both figures side by side rather than only the more favorable one.

6.3 Interpretation and what the design does not deliver

Coefficients on the index and the racial shares are on a 0–1 scale, so a unit change spans the entire possible range of the variable and is not a meaningful comparison. Throughout we scale to a ten-percentage-point (0.1) change, which is roughly two-thirds of a within-sample standard deviation of the index. For the logit we report coefficients in log-odds units, which should be scaled the same way.

Identification comes from within-district variation over time, conditional on statewide year shocks. District fixed effects absorb permanent differences in political preferences, tax capacity, institutional practice and baseline demographics; year effects absorb statewide shocks including House Bill 3, the 2019 disclosure requirement, rate compression and the pandemic. What they do not absorb is time-varying residential sorting. If families sort into or out of districts in response to expected school investment, the index is endogenous in a way fixed effects cannot address, and the three closest studies all deploy an instrument or a control group precisely for this reason. We have neither, and we describe our estimates as associations throughout.

A second limitation is selection into holding an election at all. Every row in our data is a bond proposition, so we estimate passage conditional on a district having decided to ask. Hopkins (2009) shows that Texas districts undergoing diversification are 2.1 points *less* likely to hold a rollback vote, so the selection margin is documented in our own setting rather than hypothetical, and it runs in a direction that would attenuate our estimates if districts anticipating defeat decline to ask. Balsdon et al. (2003) estimate a three-equation system — proposed spending, vote, and selection — for California school bond elections, and Romer et al. (1992) avoid the problem only because New York required annual referenda. Correcting for it would require constructing the full universe of Texas district-years, including the years no election was held, and an exclusion restriction that predicts calling an election without directly affecting passage. We have not built that panel, and Section 9 sets out what it would take.

7 Results

7.1 Main estimates

Table 4 reports equation (6.1) under the three estimators. The coefficient on the hybrid fragmentation index is -1.362 in the full-sample linear model, with a district-clustered standard error of 0.603

and $p = 0.024$. Scaled to a ten-percentage-point increase in the index — about two-thirds of a within-sample standard deviation — this implies a 13.6-percentage-point reduction in the probability that a proposition passes, against a baseline passage rate of 65.6%. The sign is negative in all three columns. In the logit the index is -16.57 in log-odds units ($p = 0.082$), and on the sample-matched linear model -1.973 ($p = 0.078$).

The pattern across the three columns is itself informative, and it is why we report the matched column at all. The point estimate grows in magnitude as the sample narrows, from -1.36 to -1.97, while the standard error grows faster, so conventional significance is lost. The honest summary is that the full-sample result is partly carried by districts whose propositions never vary in outcome: those districts contribute to the within-district demeaning that identifies the linear model, but they are dropped entirely by the logit for want of likelihood variation. The effect is not an artifact — the sign is stable and the magnitude if anything increases — but the $p < 0.05$ framing of the headline is more fragile to sample composition than a single p-value suggests.

Among the controls, the ballot-mechanics variables are the most robust relationships in the model, and all four track the literature. A May (off-cycle) election raises passage probability by 14.4 points, first position on the ballot by 19.9 points, and each additional attempt by 20.7 points, all with $p < 0.001$. Recent enrollment growth raises passage (0.224, $p = 0.003$), consistent with the standard reading that visible capacity pressure gives voters a concrete reason to approve. Bonds whose stated purpose is Building (-0.062, $p = 0.018$) or Renovations (-0.173, $p = 0.016$) are less likely to pass than the Other category. Bond size, the number of propositions, the economically disadvantaged share and median household income are all insignificant, the first two with the expected negative sign.

7.2 Comparing the three indices

The framework in Section 3 predicts that a two-population measure should outperform either within-population measure, and Table 5 tests this directly by substituting each index into the identical specification. The hybrid index is significant at conventional levels (-1.362, $p = 0.024$); adult fractionalization is not (-0.956, $p = 0.122$); student fractionalization is not (-0.871, $p = 0.154$). All three are negative, so this is a difference in precision and magnitude rather than in sign, but the ordering is the one the framework implies.

Column (4) enters both within-population indices together. Neither is significant ($p = 0.20$ in both cases), and the pair does not recover what the single hybrid index delivers. This is the paper's central measurement result: a measure defined across both populations carries information that measures defined within each population, separately or together, do not.

The hybrid index can be high for two reasons: because each population is internally diverse, or because the two populations are made up of different groups. Because the index is closely correlated

Table 4: Voter–Student Mismatch and Bond Passage: Main Estimates

Variable	(1) LPM, full sample	(2) Fixed-effects logit	(3) LPM, logit sample
Panel A: Fragmentation			
Hybrid fragmentation index, HFI	-1.3624^{**}	-16.5696^{*}	-1.9726^{*}
	(0.6032)	(9.5427)	(1.1145)
Panel B: Adult racial composition (Other omitted)			
White share, CVAP	-1.3542	-22.5516	-2.6867
	(1.5215)	(20.3691)	(2.8920)
Black share, CVAP	0.2803	-0.8775	0.0533
	(1.6894)	(21.7378)	(3.1543)
Hispanic share, CVAP	-1.9716	-25.9983	-3.1863
	(1.6312)	(22.5442)	(3.0994)
Panel C: Age structure and tax burden, with time interactions			
Population 65 and over	0.2618	2.5599	0.7961
	(1.0613)	(11.7283)	(1.7787)
$\times$ time trend	0.0642	1.3690	0.1334
	(0.0845)	(1.1284)	(0.1568)
I&S tax rate	0.1477	2.9062	0.1688
	(0.3691)	(5.1297)	(0.6494)
$\times$ time trend	-0.1266^{***}	-1.6884^{***}	-0.2189^{***}
	(0.0397)	(0.5443)	(0.0655)
Panel D: Bond and ballot characteristics			
Log bond amount per CVAP	-0.0083	-0.1143	-0.0148
	(0.0124)	(0.1273)	(0.0182)
May (off-cycle) election	0.1437^{***}	1.1856^{***}	0.1814^{***}
	(0.0346)	(0.3081)	(0.0443)
Propositions on the ballot	-0.0263	-0.1998	-0.0151
	(0.0196)	(0.2054)	(0.0276)
First on ballot	0.1991^{***}	2.0689^{***}	0.3247^{***}
	(0.0297)	(0.3081)	(0.0429)
Attempt number	0.2072^{***}	1.5981^{***}	0.2335^{***}
	(0.0332)	(0.3288)	(0.0387)
Panel E: District characteristics			
Enrollment growth, one year	0.2236^{***}	2.1394^{**}	0.2917^{***}
	(0.0756)	(0.8592)	(0.0871)
Economically disadvantaged share	0.1501	-0.4254	0.0116
	(0.4288)	(5.0505)	(0.6702)
Median household income	0.0000	0.0001	0.0000^{*}
	(0.0000)	(0.0000)	(0.0000)
Purpose: Building (vs. Other)	-0.0622^{**}	-0.5247^{**}	-0.0828^{**}
	(0.0263)	(0.2371)	(0.0384)
Purpose: Renovations (vs. Other)	-0.1726^{**}	-1.4298^{*}	-0.2751^{***}
	(0.0713)	(0.7322)	(0.1041)
Purpose: Programs (vs. Other)	-0.2365	-2.5998^{**}	-0.3858^{*}
	(0.1908)	(1.1654)	(0.2162)
District fixed effects	Yes	Yes	Yes
Election-year fixed effects	Yes	Yes	Yes
Clustered by district	Yes	Yes	Yes
Propositions	1,941	1,209	1,209
Districts	495	254	254
R^2 / pseudo-R^2	0.509	0.309	0.339

Notes: The dependent variable indicates whether the proposition passed; standard errors clustered by district in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Column (2) is a conditional logit, which drops districts whose outcome never varies; column (3) re-estimates the linear model on exactly those retained rows, so (1) versus (3) is sample composition and (2) versus (3) is functional form. A time-trend main effect and a campus-charter indicator enter every specification and are absorbed by the year and district fixed effects. Index and share coefficients are on a 0–1 scale, so a ten-percentage-point change is one-tenth of the reported value.

with both within-population indices (0.87 with adult and 0.83 with student fractionalization), these two sources are hard to separate in our data. Table 5 therefore shows that a measure built from both populations predicts passage better than measures built from either one, but these data cannot say how much of that advantage comes from the mismatch between voters and students as such.

Table 5: Which Fragmentation Measure Predicts Passage?

Index entered	**(1) Hybrid**	**(2) Adult only**	**(3) Student only**	**(4) Both within**
Hybrid fragmentation index, HFI	-1.3624^{**} (0.6032)	—	—	—
Adult fractionalization, FI^c	—	-0.9558 (0.6167)	—	-0.7920 (0.6176)
Student fractionalization, FI^s	—	—	-0.8707 (0.6099)	-0.7704 (0.6038)
p-value on index	0.024	0.122	0.154	0.200 / 0.203
Effect of a $+0.10$ change (pp)	-13.6	-9.6	-8.7	—
Full control set	Yes	Yes	Yes	Yes
District, year fixed effects	Yes	Yes	Yes	Yes
Propositions	1,941	1,941	1,941	1,941
R^2	0.509	0.507	0.508	0.508

Notes: Each column is the specification of Table 4, column (1), with the fragmentation measure replaced as indicated and everything else held identical, including the adult racial shares, the full control set, district and election-year fixed effects, and district-clustered standard errors. $^{***}\,p<0.01$, $^{**}\,p<0.05$, $^{*}\,p<0.10$. In column (4) the two p-values correspond to FI^c and FI^s respectively. The effect row scales the coefficient to a ten-percentage-point increase in the index and expresses it in percentage points of passage probability.

7.3 Two competing time patterns: age and tax burden

The specification in Table 4 contains two time interactions, and their relationship to one another is the second substantive finding of the paper. Table 6 reports the two nested specifications side by side, with sample size held identical at 1,941 so that nothing in the comparison is driven by composition.

Begin with what the literature would lead one to expect. Bowers and Lee (2013) find the elderly share negatively related to bond passage in Texas over 1997–2009. Our pooled estimate, without any time interaction, is positive and insignificant. Investigating this was the original motivation for the time interaction, and the answer appeared clean: in Model A, the elderly share interacted with a linear trend is positive and significant (0.186, $p = 0.019$), with the main effect negative, implying a relationship that began the period negative — matching Bowers and Lee — and moved upward across it. Split-sample estimates point the same way. The Texas school tax ceiling for homeowners aged 65 and over, which has been made progressively more generous since 2007 (Section 4), offers a plausible mechanism: an elderly Texas homeowner's exposure to a new bond has weakened over time.

Model B undoes this. Once the I&S tax rate is allowed its own interaction with the same trend, the age interaction falls to 0.064 and becomes insignificant ($p = 0.448$), while the tax interaction

is -0.127 with $p = 0.002$. The pattern holds across all three estimators: the age interaction is significant at $p = 0.019$, 0.015 and 0.031 when the tax interaction is absent, and at $p = 0.448$, 0.225 and 0.396 when it is present, with N held fixed. A step-by-step diagnostic confirms that neither the sample restriction nor the racial shares are responsible; the tax interaction specifically absorbs it.

The interpretation of the tax variable also changes between the two columns. Without an interaction, a district's existing debt-service rate carries a large and highly significant constant penalty (-0.768, $p = 0.004$). With the interaction, that constant penalty vanishes entirely (0.148, $p = 0.689$) and the negative time interaction takes its place. Read in the correct units — recall that this variable is a continuous tax rate, not an election-type indicator — Model B says that the penalty a district pays at the ballot box for already carrying a high debt-service rate has become more negative across 2014–2025. Voters in already heavily taxed districts have become less willing to approve new borrowing. This is the most robust relationship in the entire model: negative with $p < 0.003$ in all nine specifications we estimate.

The two stories are not separately identified. Both interactions are built on the same linear trend and both are candidate explanations for the same underlying drift in how districts vote across the period. The data prefer the tax story — it survives the inclusion of the age story, and not the reverse — but the model cannot rule out that the age relationship is real and merely correlated with it. Following our own prior practice, we disclose this rather than resolve it by choosing a favorite, and we report both columns in the main text.

Throughout, the fragmentation index is stable. It is negative in all six estimates in Table 6, significant in all three Model A variants ($p = 0.006$, 0.039, 0.038), and in Model B significant in the full-sample linear model ($p = 0.024$) and marginal in the two smaller-sample estimators ($p = 0.082$, 0.078). Its sign never changes and its magnitude moves only modestly. The choice between Model A and Model B is consequential for the age and tax coefficients and inconsequential for the index.

7.4 Is the tax result composition or behavior?

A natural objection to the tax finding is that it may reflect a change in *which* districts hold elections rather than a change in how voters respond. If districts carrying high debt-service rates became more common among election-holders over the period, a growing penalty would appear in the data without any voter having changed their mind. Figure 2 and the following decomposition address this directly.

The mean I&S rate among districts holding elections did rise, from \$0.212 per \$100 in 2014–2019 to \$0.271 in 2020–2025, a difference of about six cents that is highly significant ($t = 8.50$). Regressed on the calendar year, the annual series rises by \$0.0079 per year ($p < 0.001$, $R^2 = 0.72$). Taken alone this supports the composition story.

It does not survive the within-district comparison. Regressing the I&S rate on the time trend

Table 6: Age Composition Versus Tax Burden: Two Readings of the Same Drift

Variable	Model A: no tax × trend			Model B: with tax × trend		
	LPM full	**Logit**	**LPM matched**	**LPM full**	**Logit**	**LPM matched**
Hybrid fragmentation index	-1.641^{***}	-17.79^{**}	-2.338^{**}	-1.362^{**}	-16.57^{*}	-1.973^{*}
	(0.597)	(8.62)	(1.123)	(0.603)	(9.54)	(1.114)
Population 65 and over	-0.636	-8.64	-0.848	0.262	2.56	0.796
	(1.047)	(10.77)	(1.768)	(1.061)	(11.73)	(1.779)
× time trend	0.186^{**}	2.558^{**}	0.329^{**}	0.064	1.369	0.133
	(0.079)	(1.054)	(0.152)	(0.085)	(1.128)	(0.157)
I&S tax rate	-0.768^{***}	-9.81^{***}	-1.466^{***}	0.148	2.91	0.169
	(0.264)	(3.10)	(0.436)	(0.369)	(5.13)	(0.649)
× time trend		*not in model*		-0.127^{***}	-1.688^{***}	-0.219^{***}
				(0.040)	(0.544)	(0.066)
p on age × trend	0.019	0.015	0.031	0.448	0.225	0.396
p on tax rate	0.004	0.002	0.001	0.689	0.571	0.795
Propositions	1,941	1,209	1,209	1,941	1,209	1,209

Notes: Models A and B are identical except that B adds an interaction between the district's I&S tax rate and the linear time trend; the tax rate's main effect is present in both. Sample sizes are identical within estimator, so the comparison is not driven by composition. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. The I&S tax rate is the district's continuous debt-service property tax rate in dollars per \$100 of assessed valuation, *not* an indicator for a tax-rate election. The time trend is a linear counter beginning at zero in the sample's first election year; its main effect is absorbed by the year fixed effects. Recoding the trend as the raw calendar year leaves the interaction coefficient numerically identical and changes only the uninterpretable main effect, as it must.

with district fixed effects gives \$0.0059 per year ($p < 0.001$) — roughly three-quarters of the unconditional \$0.0079. The dominant source of the rise is therefore the *same* districts accumulating debt-service obligations over time, not a change in the composition of who asks. About a quarter of the increase is compositional, which is not nothing and which we report as a qualification rather than dismissing.

Two implications follow. The behavioral reading of the tax interaction — that voters have become more sensitive to the debt burden a district already carries — is the better-supported one, and we adopt it. But because both the tax rate and the passage rate trend over the period, the interaction is partly identified from two variables moving in opposite directions over time, and we would not want to rest a strong causal claim on it. Year fixed effects absorb the common time path of passage, so the interaction is identified from *differential* passage by tax level across years rather than from the trend itself, which is the right comparison; but the concentration of elections in the later years (1,362 of 1,941 propositions fall in 2020–2025) means the later period does most of the work.

8 Robustness

This section reports six checks of the main result.

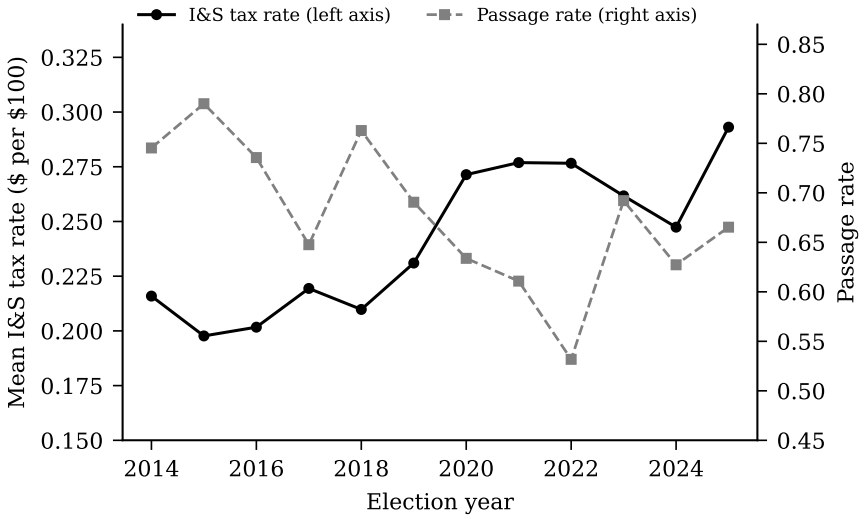


Figure 2: Debt-service tax burden and passage over the sample period

Notes: Annual means across the estimation sample. The left axis shows the mean I&S (debt-service) tax rate among districts holding bond elections that year; the right axis shows the share of propositions that passed. The I&S series rises by \$0.0079 per year unconditionally and by \$0.0059 per year within districts, so roughly three-quarters of the increase reflects the same districts accumulating debt service rather than a change in which districts hold elections.

8.1 Level versus change in the index

Hopkins (2009) finds, in Massachusetts municipalities and Texas school districts, that diversity reduces willingness to raise taxes where demographic change has been *sudden*. This motivates asking whether the change in the index, and not only its level, is related to bond passage.

Table 7 tests this by constructing the year-over-year and three-year change in the index from the full district–year panel — including years in which no election was held, so the differences are true calendar lags rather than differences across irregular gaps — and entering them alongside the level.

In the one-year specification, the level survives essentially untouched (-1.352, $p = 0.038$ with the change included, against -1.362, $p = 0.024$ without), while the one-year change is not significant (-0.058, $p = 0.958$). Entered on its own the change is also insignificant (-0.746, $p = 0.465$).

The three-year specification is less decisive and we report it as such. Requiring three prior years costs 253 observations, and on that reduced sample the level itself is no longer significant (-1.061, $p = 0.171$) even before the change term is added. The three-year change is insignificant ($p = 0.579$ alone, $p = 0.830$ alongside the level), but since the level does not survive the sample restriction either, this column speaks to power more than to substance.

These results should not be read as showing that change in racial composition is unrelated to bond passage. Racial composition moves little within a single year in either population — the average absolute one-year change in fractionalization is about 0.016 for adults and for students alike, and 0.014 for the index — and on the adult side consecutive CVAP vintages also share four of

their five survey years. The three-year version costs too much of the sample to be informative. A test over a longer horizon would be more informative, and it should separate the two populations, since the argument in Hopkins (2009) concerns change in who lives in a place, which here is the adult side. We leave for future work a comparison of three versions — change in adult composition, change in student composition, and change in the index — which would show which population's change, if either, matters.

Table 7: Level Versus Change in the Fragmentation Index

	One-year change			Three-year change		
Variable	**(1) Level**	**(2) Change**	**(3) Both**	**(4) Level**	**(5) Change**	**(6) Both**
Level, HFI_{t-1}	-1.3624^{**} (0.6032)	—	-1.3520^{**} (0.6486)	-1.0614 (0.7732)	—	-1.1576 (0.9234)
Change, $\Delta_1 HFI$	—	-0.7455 (1.0186)	-0.0577 (1.0862)	—	—	—
Change, $\Delta_3 HFI$	—	—	—	—	-0.4580 (0.8254)	0.2080 (0.9697)
p on the level	0.024	—	0.038	0.171	—	0.211
p on the change	—	0.465	0.958	—	0.579	0.830
Full control set	Yes	Yes	Yes	Yes	Yes	Yes
District, year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Propositions	1,941	1,941	1,941	1,688	1,688	1,688

Notes: Changes are computed on the complete district–year panel, including years with no bond election, so $\Delta_1 HFI_{d,t-1} = HFI_{d,t-1} - HFI_{d,t-2}$ is a true one-year difference rather than a difference across an irregular gap. All columns otherwise replicate Table 4, column (1). $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Columns (4)–(6) lose 253 propositions to the three-year lag requirement; note that the level is itself insignificant on that reduced sample before any change term is added, so those columns are uninformative about the level-versus-change question. The sample mean of $\Delta_1 HFI$ is $+0.005$ with a standard deviation of 0.017.

8.2 Vote share as an alternative outcome

Passage is a threshold outcome, and it is natural to ask whether the index also predicts the intensity of support. Re-estimating equation (6.1) with the yes-vote share as the dependent variable, the index coefficient is -0.088 with a standard error of 0.169 ($p = 0.604$). Restricted to the rows the logit retains, it is -0.221 with a standard error of 0.265 ($p = 0.404$). Neither is close to conventional significance.

This is an honest negative and we do not want to soften it. Several controls behave exactly as they do in the passage model — May elections, first ballot position and attempt number are all significant with the same signs, and the tax-rate time interaction remains significant (-0.033, $p = 0.003$) — so the vote-share specification is not simply underpowered or broken. The index specifically does not carry over.

We can offer three readings and cannot distinguish among them. It may be that the relationship operates near the decision threshold rather than across the whole distribution of support, which would be consistent with a story in which mismatch matters for marginal turnout and mobilization

rather than for the average voter's preference. It may be that vote share is noisier in a way that particularly affects a slow-moving district-level regressor, since vote share in our data includes some extreme values including zeros and ones. Or it may be that the passage result is less robust than it appears, and this is the check that reveals it. Readers should weigh the passage result in light of this.

8.3 Subperiods

Splitting the sample at 2019 — the year of the ballot-disclosure requirement and roughly the midpoint of the period — and estimating without time interactions, the index is $+0.552$ ($p = 0.814$) in 2014–2019 and -3.862 ($p = 0.008$) in 2020–2025. The estimated relationship comes almost entirely from the later period.

Two competing explanations deserve mention, and the data cannot separate them. The early subsample is small: only 372 propositions survive the fixed-effect requirements, against 1,275 in the later period, because bond activity in Texas rose sharply after 2020. Low power alone could produce this pattern: with a standard error of 2.34, the positive early estimate is fully consistent with a negative effect of the size estimated for the later period. Alternatively the relationship may genuinely have strengthened, in which case the pooled estimate understates what is happening now. We report the split because a reader should know the result is not uniform across the period, and we decline to interpret the early-period point estimate given its standard error of 2.34.

8.4 Charter proximity and campus charters

Open-enrollment charter schools have no bond authority and never appear in the election file, but their presence nearby could plausibly affect how a district's voters view school investment. We construct the straight-line distance from each district to the nearest open-enrollment charter campus, using the zip-code centroids of the district's address and of each charter campus in the TEA school directory, flag districts within ten and fifteen miles, and interact each flag with the index.

Neither interaction is significant. At ten miles the interaction is $+0.782$ ($p = 0.423$) with the index itself remaining significant (-1.785, $p = 0.028$); at fifteen miles the interaction is -0.516 ($p = 0.605$) and the index loses significance when split by that flag (-1.025, $p = 0.233$). A third specification interacting the index with an indicator for the district hosting its own campus charter gives a large but very imprecise negative interaction (-3.119, $p = 0.429$ in the linear model; -52.14, $p = 0.139$ in the logit). Only thirteen districts in the estimation sample host a campus charter, contributing 73 propositions, so this specification has little power and we build nothing on it.

Two limitations apply to this check. Distances are computed from zip-code centroids rather than street addresses, and they use current charter locations as a stand-in for each past election year, since a historical charter map is not available. Separately, a campus-charter enrollment share built from seven years of archived TEA directory files (2018–2024, the full extent of the public archive)

moves the index coefficient only trivially when added on the matched sub-sample, from -1.136 to -1.095. The charter-campus share is itself positive and significant in the linear model on that sub-sample, which is a finding we note but do not pursue.

8.5 Outliers and data quality

One observation lies well outside the plausible range: a district–election date recording one-year enrollment growth of 286%, which appears twice because two propositions shared that ballot. Dropping it gives -1.240 ($p = 0.044$). The result is not driven by it, though it does move the point estimate by about nine percent.

8.6 Sample definition and exclusions

Table 8 re-estimates the main specification of Table 4, column (1), on alternative samples. Four checks drop observations that might behave differently from the rest: refunding bonds, which replace existing debt rather than finance new capital; districts in the 17 counties of the Texas–Mexico border region; the smallest tenth of districts by CVAP; and the smallest tenth of propositions by amount per CVAP. In each case the index remains negative and significant at the 5% level, with coefficients between -1.20 and -1.64. Requiring each district to have held elections on at least two distinct dates, rather than to have at least two propositions, removes the 72 districts whose fixed effect would otherwise be identified only from propositions on a single ballot, and leaves the estimate essentially unchanged (-1.386, $p = 0.022$).

Two checks concern the 2020 change in census tract boundaries (Section 5.2). The land-area tract-to-district crosswalk, used as an alternative to the population-weighted crosswalk in earlier work, gives a larger and more precisely estimated coefficient (-1.553, $p = 0.011$). It has been dropped from routine use because its name-based matching cannot disambiguate the twelve district names shared by two real Texas districts, and it retains 60 fewer propositions. The second check drops the districts whose adult fractionalization jumped between 2019 and 2020 by more than 0.02 more under population weighting than under land-area weighting — the pattern expected if the crosswalk switch, rather than real demographic change, drove the jump. Thirty of the 49 flagged districts appear in the estimation sample. Dropping them reduces the index to -1.039 ($p = 0.095$). This is the weakest result in the table: the sign and approximate magnitude hold, but the estimate is no longer significant at the 5% level, and we report it as such.

9 Discussion

9.1 What the index is measuring

The result we set out to test was whether a measure of racial distance between the electorate and the student body predicts school bond passage better than conventional within-population

Table 8: Alternative Samples

Sample	HFI (s.e.)	p-value	Propositions	Districts
Main estimation sample	−1.3624 (0.6032)	0.024	1,941	495
Excluding refunding bonds	−1.3990 (0.6109)	0.022	1,908	488
Excluding border-region districts	−1.2023 (0.6086)	0.049	1,876	472
Excluding smallest 10% of districts (CVAP)	−1.5289 (0.6621)	0.021	1,806	451
Excluding smallest 10% of bonds (amount per CVAP)	−1.6390 (0.5941)	0.006	1,736	489
Districts with two or more election dates	−1.3863 (0.6040)	0.022	1,772	423
Land-area crosswalk	−1.5526 (0.6048)	0.011	1,881	479
Excluding districts with large 2019–20 crosswalk jumps	−1.0394 (0.6214)	0.095	1,846	467

Notes: Each row re-estimates Table 4, column (1), on the sample indicated, with the full control set, district and election-year fixed effects, and standard errors clustered by district; fixed-effect singletons are dropped after each restriction. Border-region districts are those in Brewster, Brooks, Cameron, Dimmit, El Paso, Hidalgo, Hudspeth, Jim Hogg, Kinney, Maverick, Presidio, Starr, Terrell, Val Verde, Webb, Zapata and Zavala counties. Deciles are computed on the complete-case sample. The land-area row replaces the index and the adult racial shares with their land-area-weighted counterparts. The last row drops the 30 sample districts among the 49 whose 2019–2020 jump in adult fractionalization exceeds the land-area jump by more than 0.02.

diversity measures. It does: the hybrid index is significant at $p = 0.024$, while adult and student fractionalization reach only $p = 0.12$ and $p = 0.15$ in the identical specification, and the two together do not recover what the hybrid index delivers.

What the comparison cannot tell us is why. The index can be high because each population is internally diverse or because the two populations are made up of different groups, and only the second is mismatch in the sense of the framework. Our data do not separate the two: the index correlates 0.87 with adult and 0.83 with student fractionalization. Two interpretations are therefore consistent with our data. In the first, voters respond to the distance between themselves and the students, as the framework assumes. In the second, the index predicts passage because of the way it combines the diversity of the two populations, which tracks bond outcomes better than either fractionalization index alone or both entered side by side. Both readings imply that when a public good is financed by one population and used by another, a summary of racial structure should be built from both populations.

The distinction matters for how the measure is described in future work. Calling the HFI a measure of voter–student distance is accurate as a matter of construction — it is built from the shares of both populations — but our estimates should be read as evidence that this summary predicts passage, not as proof that mismatch, as distinct from diversity, changes how people vote. Telling the two apart would require variation in the mismatch between voters and students that is separate from the diversity of each population, for example from a design that can credibly use differences

across districts.

9.2 Why the tax result may be the more important one

The most robust relationship we document is not demographic. A district's existing debt-service tax rate carries a penalty that grows across 2014–2025, negative and significant at the 1% level in every one of the nine specifications we estimate. Among the demographic and fiscal variables it is the only one with that property; the others that meet it are features of the ballot itself — a May election date, first position on the ballot, and the attempt number. It also absorbs an age-composition pattern that looks significant in its absence, which is a cautionary result for a literature that has concentrated on demographics.

Section 7.4 shows that about three-quarters of the rise in debt-service rates among election-holding districts is within-district accumulation rather than a change in who asks, which supports the behavioral reading: voters have become less willing to add to a burden they already carry. That reading is consistent with the tax-salience literature (Chetty et al., 2009; Finkelstein, 2009; Cabral and Hoxby, 2012), and particularly with Brunner et al. (2018) and Brunner et al. (2021), who show that making the property tax consequence explicit on the ballot reduces support substantially. Texas imposed exactly such a disclosure requirement in 2019, in the middle of our window, and Heller and Lyon (2026) estimate its effect at about nine percentage points using a control group we do not have.

The natural hypothesis — that the growing tax penalty reflects rising salience of debt service after 2019 — is one our design cannot test, and the evidence we have does not discriminate between a gradual trend and a discrete break. Interacting the tax rate with an indicator for elections in 2019 or later, instead of a linear trend, also yields a large, significant negative coefficient (-0.743, $p = 0.010$ in the linear model; -9.69, $p = 0.010$ in the logit). When both interactions are entered together, neither is cleanly identified: the trend term retains most of the magnitude (-0.128, $p = 0.102$) while the break term collapses to zero (0.009, $p = 0.987$). That pattern depends on where the break is placed. The disclosure requirement took effect on September 1, 2019 and did not apply to any 2019 election,[2] so the 2019-onward indicator counts a year of untreated elections as treated. If the break is dated to 2020 instead, the result reverses: the trend term falls to zero and the break term carries the effect (-0.975, $p = 0.065$). Two highly collinear representations of the same change cannot be separated in these data. We adopt the linear trend in the reported specifications because it does not require choosing a break date; for the racial shares, the interaction with the Other adult share is significant with a linear trend in both the linear and logit models but with a 2019-onward

[2] Office of the Texas Attorney General, Public Finance Division, letter to bond counsel of July 19, 2019, Exhibit A, which lists the House Bill 3 ballot-language requirement as effective September 1, 2019 and not applicable to the November 5, 2019 election.

indicator only in the logit. We do not claim to have distinguished the two, and a reader who prefers the disclosure interpretation will find our data consistent with it. We flag the connection to Heller and Lyon (2026) as the most promising direction for follow-up rather than as a finding.

9.3 The elderly, and what our estimate does and does not say

Our pooled elderly coefficient is positive and insignificant, against Bowers and Lee (2013)'s significant negative estimate in the same state for 1997–2009. Three points are worth making.

First, a small district-level elderly effect is the documented expectation. Harris et al. (2001) estimate a district-level elasticity about 40% of Poterba's state-level figure; Brunner and Balsdon (2004) show in survey data that, although support declines with age, older voters are more willing to back local than state school spending; Ladd and Murray (2001) find no within-county effect at all. Our setting is purely local, which is where the elderly are least oppositional.

Second, our measure cannot distinguish the groups that matter. Berkman and Plutzer (2004) show across more than 9,000 districts that longstanding older residents support school spending while elderly in-migrants depress it. A single 65-and-over share averages over both and will be attenuated wherever they coexist. This is a measurement limitation of our data, not a finding.

Third, the apparent time pattern is not separately identified from the tax pattern, as Section 7.3 documents at length. We report both and do not claim the age story.

The Texas tax ceiling for homeowners aged 65 and over remains an attractive explanation for why elderly Texans might be less oppositional than elderly voters elsewhere, since their exposure to a new bond is capped and has been capped more generously over time. But it cannot explain a change within Texas, because the ceiling was already in force throughout Bowers and Lee's window. We name it as a hypothesis and test it nowhere.

9.4 Limitations

No causal design. We have neither an instrument nor a control group. Alvord and Rauscher (2021) instrument with national demographic trends, Jones and Zhan (2020) use a shift-share instrument, and Brunner et al. (2018) have a neighboring state. Our estimates are within-district associations. The most likely direction of bias is through time-varying residential sorting, which district fixed effects do not absorb: if families sort in response to expected school investment, the index is endogenous. Notably, when Jones and Zhan instrument, their negative diversity coefficient falls to less than a quarter of its OLS value in their main specification and loses significance — a direct warning about what an instrument might do to ours. Building the shift-share analogue for Texas, following Boustan et al. (2013) as Jones and Zhan (2020) do, is the single most valuable extension available.

Selection into holding an election. We observe only elections that occurred. Hopkins (2009) shows that diversifying Texas districts are 2.1 points less likely to hold a rollback tax vote, so the margin is documented in our own setting. Balsdon et al. (2003) estimate a three-equation system — proposed spending, vote, and selection — for California school bond elections and find school boards behave like risk-averse, budget-maximizing agenda setters. Closing this requires constructing the full universe of Texas district-years and an exclusion restriction that predicts calling an election without directly affecting passage; plausible candidates include time since the last bond and statutory election-calendar windows. We have not built that panel. Balsdon, Brunner and Rueben show that the two decisions can be modeled jointly in this setting; doing so for Texas is the natural next step.

The index cannot say who. An index of total distance between two distributions is agnostic about which pairing drives it. Brunner and Johnson (2016) can state that older white voters withdraw support for community-college bonds where the college-age population is more heavily Hispanic; we cannot. This is the price of generality, and it is worth paying only if the general measure earns its place. Section 9.1 shows that it predicts passage better than either within-population measure, but not yet that it does so because of the gap between voters and students.

Functional form of the index. Montalvo and Reynal-Querol (2005) argue that fractionalization is the wrong functional form for questions about group conflict, since fractionalization rises monotonically with the number of groups while polarization peaks when two comparably sized groups face each other. A referee may reasonably ask whether a polarization-type measure suits a two-population comparison better. Jones and Zhan (2020) repeat their analysis with a Reynal-Querol polarization index and obtain the same pattern they find with fractionalization, which is mild evidence that the choice may not be decisive. We have not run the analogue.

Measurement. Our four-group scheme folds Asian residents and students into an Other category, a decision forced by cell sizes (43% of the propositions in our panel come from district-years with zero Asian students) but one that discards real variation in metropolitan districts. CVAP arrives in overlapping five-year vintages, so our annual variation is smoothed. Charter distances use current locations and zip-code centroids. The attempt-number variable is a heuristic and a lower bound on true resubmissions. Beach and Jones (2017) raise a further possibility our design cannot exclude: if diversity operates through the composition of the governing body rather than the electorate, part of any index effect could run through school board behavior — boards choose bond size, timing and purpose — rather than through voters.

Unit of observation. Our unit is the proposition, which matches Bowers and Lee (2013), but propositions sharing a ballot face the same electorate on the same day. District clustering addresses

the resulting dependence, but a specification collapsing to the district–election date would be a reasonable alternative and we have not estimated it.

10 Conclusion

School bonds separate the people who pay from the people who benefit more sharply than almost any other public finance decision made by voters. This paper asked whether the racial distance between those two populations predicts whether a bond passes, using Texas independent school district bond propositions from 2014 through 2025 and measuring both populations annually — the electorate from the citizen voting-age population, the beneficiaries from state enrollment records.

The answer has two parts.

The first part is that a two-population index outperforms conventional within-population measures. A ten-percentage-point increase in the hybrid fragmentation index is associated with a 13.6-percentage-point lower probability of passage ($p = 0.024$), while adult fractionalization ($p = 0.12$) and student fractionalization ($p = 0.15$) are individually insignificant in the identical specification, and remain so when entered together. For a literature that has measured diversity within one population at a time — and, in Bowers and Lee (2013)'s study of Texas bond propositions, has held adult demographics fixed at a decennial census across a thirteen-year panel — this is a concrete improvement in how racial structure is summarized.

The second part is a qualification about what drives this. The index can be high because each population is diverse or because the two populations are made up of different groups, and only the second is mismatch as the framework intends, and our data cannot separate the two. Our results therefore show that a summary of racial structure built from both populations predicts passage; they do not establish that mismatch, as distinct from diversity, changes how people vote.

Two further results stand. The index predicts passage but not vote share, an asymmetry we report without being able to explain it. And the most robust relationship in the data is not demographic at all: the penalty attached to a district's existing debt-service tax rate grows across the period, is significant at the 1% level in every specification, and absorbs an age-composition pattern that appears significant when it is omitted. Roughly three-quarters of the underlying rise in debt-service rates is within-district accumulation rather than a change in which districts hold elections, so the behavioral reading is the better supported one.

None of these are causal estimates. We have no instrument and no control group, and we observe only the elections districts chose to hold — a margin Hopkins (2009) shows is itself related to diversification in Texas. The most valuable extensions are correspondingly clear: a shift-share instrument for the index following Jones and Zhan (2020); a test of whether longer-horizon change matters, as Hopkins (2009)'s argument about sudden demographic change would predict, comparing change in adult composition, change in student composition and change in the index; and a selection

model built on the full universe of Texas district-years, which would let the two questions a district faces — whether to ask, and whether voters agree — be estimated jointly rather than one conditional on the other. The last is a data-construction task more than a methodological one, and it is where we intend to go next.

For policy, the durable finding is the least novel-sounding one. Districts already carrying heavy debt service face a penalty at the ballot box that has grown over the past decade, and that penalty is more robust across specifications than anything we can attribute to demographic composition. Whatever the mechanism connecting who votes to who benefits, among the characteristics of a district rather than of the ballot, its accumulated tax burden is the one that most reliably predicts whether its voters will approve more.

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A Data Construction and Merge Diagnostics

This appendix documents the geographic and temporal matching behind the district–year panel, together with the match rates and the known gaps.

A.1 Geography chain

Adult composition begins at the census tract. The Census Bureau's CVAP Special Tabulation reports citizen voting-age counts by race and ethnicity for each tract, in fourteen overlapping five-year ACS vintages from 2007–2011 through 2020–2024. Tracts are allocated to school districts using the Missouri Census Data Center's *geocorr* population-weighted crosswalks: the 2018 vintage for years through 2019 and the 2022 vintage from 2020 onward. The handoff between vintages coincides exactly with the Census Bureau's decennial tract redraw, visible in the data as a jump in Texas tract count from 5,265 to 6,896 between the 2015–2019 and 2016–2020 CVAP vintages. Between 99.96% and 100% of tract-level CVAP totals survive allocation in every year and under both crosswalks.

Tract boundaries are not stable, and this is a genuine limitation rather than a technicality. The Bureau targets roughly 4,000 residents per tract, so fast-growing areas — outer Houston, the Dallas–Fort Worth suburbs, the Austin exurbs — have tracts split between vintages, and the whole state is redrawn each decade. Our pipeline handles this with a single switch rather than a gradual update. The switch is not innocuous. A diagnostic comparing district-level adult fractionalization either side of it flags 49 districts whose 2019–20 change is more than 0.02 larger under population weighting than under land-area weighting; 30 of them are in the estimation sample, and dropping them reduces the index coefficient to -1.04 ($p = 0.095$; Section 8.6). A separate diagnostic finds nine district codes present only in the 2010-vintage crosswalk (Byers, Higgins, Kendleton, La Marque, North Forest, Novice, Roxton, Samnorwood and Star) and six only in the 2020 vintage (Bluff Dale, Gholson, Lasara, London, McDade, and the placeholder code "99999", which denotes territory in no district). Only three of these districts — Bluff Dale, London and McDade — appear in the estimation sample, with 13 propositions between them, all from 2020 onward; dropping them leaves the index coefficient at -1.38. Whether each code change reflects a consolidation, an annexation or a change in district type has not been checked against TEA records.

A.2 Match rates

Student enrollment and economically disadvantaged counts are joined to the adult side on a cleaned district name plus county, with a fallback to name alone where unique. Cleaning uppercases the name, strips ISD/CISD suffixes, and expands common abbreviations. In the TEA district directory, twelve cleaned district names are each shared by two different Texas districts — Big Sandy, Centerville, Chapel Hill, Dawson, Edgewood, Highland Park (Dallas and Amarillo), Hubbard, Midway, Northside, Rice, Valley View and Wylie (Abilene and Collin County) — and these 24 districts rely on county to disambiguate.

- CVAP to district allocation: 99.96–100% of tract totals preserved, all fourteen years.
- Student and economically disadvantaged records to adult CVAP: 98.9% of district-years matched,

2012–2024. Twenty-five districts never match, mostly very small districts or unusual partnership entities (Texas Tech University K–12, UT Austin High School, several "Collegiate" partnership districts) that do not appear as standalone units in Census school-district geography.

- Bond elections to the district–year panel: 99.0% matched for elections from 2012 onward.
- I&S tax rate: joined on TEA's exact county-district number rather than by name, 4,829 of 4,868 rows matched.

The land-area crosswalk used in earlier rounds matches 97.3% of bond elections. It cannot disambiguate the shared names for want of a county field, and every one of the 18 districts it fails to match entirely carries one of the twelve shared names. Across 14,806 overlapping district-years the two crosswalks correlate 0.983 on adult fractionalization and 0.996 on the hybrid index, with mean absolute differences of 0.014 and 0.007. The land-area version has been dropped from routine use.

A.3 Race categories

Both populations are collapsed to White, Black, Hispanic and Other non-Hispanic, where Other combines American Indian and Alaska Native, Asian, Native Hawaiian and Pacific Islander, and two or more races. We tested a five-category scheme separating Asian residents and students, restricted to the 4,868 rows of the bond panel rather than the full universe of more than 14,000 district-years. Adult Asian CVAP has a median of roughly 45–47 persons, but 31–45% of matched rows record fewer than thirty Asian adults, and Asian residents average 1.6% of adult CVAP with a median of 0.6%. On the student side, 2,084 to 2,145 of the 4,868 rows (43%, the range reflecting the two crosswalks) come from district-years with zero Asian students. A regressor identically zero in more than two-fifths of the sample cannot support district and year fixed effects, and the four-group scheme was retained on that basis.

A.4 Derived controls

Ballot-mechanics variables are constructed from the election file. The attempt number counts election-level attempts, incrementing only when the immediately preceding election in the same district was fully defeated and occurred within 24 months; districts that pass first time or wait more than two years are coded as first attempts. Across the full file the distribution is 3,144 elections at attempt one, 373 at attempt two, 69 at three, 7 at four and 1 at five. The variable is a lower bound on true resubmissions because the source does not preserve a linked bond identifier.

Turnout as a share of CVAP exceeds one for eleven of roughly 4,825 non-missing rows, almost all in very small districts where district-level CVAP is itself an imprecise ACS estimate. One larger case, Frisco ISD in 2018 with an implied turnout of 2.49, is not root-caused; turnout is not used in any specification reported here.

A.5 Known gaps

Three items remain open and are listed so a reader can weigh them. TEA accountability ratings and district assessed value were considered as controls and deliberately not pulled: the I&S rate and assessed value are mechanically related (rate $\approx$ debt service $\div$ assessed value) and including both as levels risked collinearity for limited added information, while assessed value is available as a convenient statewide export only for the most recent year or two. Average household size (ACS table B25010) and the unemployment rate (B23025) are plausible controls used in adjacent literatures but we found no bond-passage-specific precedent for either, and neither is in the panel. The school-level extension — identifying which campuses a bond targets, so that racial composition can be measured at the level of the beneficiary school rather than the district — is outlined in earlier project documents and not started.

B Supplementary Results

B.1 Subperiod estimates

Table B1 reports the split-sample estimates discussed in Section 8.3, estimated without time interactions so that the index and the tax rate carry their own period-specific coefficients.

Table B1: Subperiod Estimates

Variable	(1) 2014–2019	(2) 2020–2025	(3) Pooled
Hybrid fragmentation index	0.5516	-3.8622^{***}	-1.2900^{**}
	(2.3356)	(1.4402)	(0.5917)
I&S tax rate	-0.5830	-1.2237^{**}	-0.7082^{***}
	(0.6302)	(0.5631)	(0.2609)
Population 65 and over	0.9349	2.7808	0.8167
	(1.9944)	(1.7144)	(0.8828)
Adult racial shares	Yes	Yes	Yes
Full control set	Yes	Yes	Yes
District, year fixed effects	Yes	Yes	Yes
Propositions	372	1,275	1,941

Notes: Time interactions are omitted in all three columns so that each period's coefficients stand on their own. $^{***}\,p<0.01$, $^{**}\,p<0.05$, $^{*}\,p<0.10$. The 2014–2019 subsample retains only 372 propositions after the fixed-effect requirements, because Texas bond activity rose sharply after 2020; the early-period point estimate has a standard error of 2.34 and should not be interpreted. Column (3) differs from Table 4 because it omits both time interactions.

B.2 Vote share as the outcome

Table B2 reports the vote-share specification discussed in Section 8.2. We show the full coefficient vector because the point of the table is that the controls behave normally while the index does not.

B.3 Charter proximity interactions

Table B3 reports the three charter interactions from Section 8.4.

Table B2: Yes-Vote Share as the Dependent Variable

Variable	(1) Full sample	(2) Logit-matched sample
Hybrid fragmentation index	−0.0875	−0.2213
	(0.1688)	(0.2648)
White share, CVAP	−0.0322	0.0781
	(0.4290)	(0.6649)
Black share, CVAP	0.2310	0.6553
	(0.4564)	(0.6641)
Hispanic share, CVAP	0.0439	−0.1294
	(0.4394)	(0.6836)
Population 65 and over	−0.0271	−0.1523
	(0.3056)	(0.4710)
× time trend	0.0253	0.0313
	(0.0234)	(0.0361)
I&S tax rate	0.0858	0.1520
	(0.1078)	(0.1569)
× time trend	−0.0326***	−0.0500***
	(0.0110)	(0.0160)
Log bond amount per CVAP	−0.0044	−0.0036
	(0.0028)	(0.0035)
May (off-cycle) election	0.0441***	0.0458***
	(0.0084)	(0.0101)
Propositions on the ballot	−0.0039	−0.0075
	(0.0044)	(0.0055)
First on ballot	0.0626***	0.0702***
	(0.0065)	(0.0080)
Attempt number	0.0295***	0.0345***
	(0.0077)	(0.0087)
Enrollment growth, one year	−0.0007	0.0078
	(0.0192)	(0.0229)
Economically disadvantaged share	−0.0575	−0.0376
	(0.1182)	(0.1776)
Median household income	0.0000	0.0000
	(0.0000)	(0.0000)
Purpose: Building (vs. Other)	−0.0115*	−0.0150*
	(0.0060)	(0.0081)
Purpose: Renovations (vs. Other)	−0.0207	−0.0447*
	(0.0187)	(0.0251)
Purpose: Programs (vs. Other)	−0.0389	−0.0576
	(0.0358)	(0.0423)
District, year fixed effects	Yes	Yes
Propositions	1,941	1,209
Districts	495	254
R^2	0.677	0.431

Notes: The dependent variable is the yes-vote share. Both columns use the main specification of Table 4 (Model B, with the elderly-share and tax-rate time interactions), with standard errors clustered by district. Column (2) restricts the sample to the propositions retained by the fixed-effects logit in Table 4, column (2). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The campus-charter indicator and the time-trend main effect are absorbed by the fixed effects. The substantive point is that the ballot-mechanics controls and the tax-rate interaction behave as they do in the passage model, while the fragmentation index does not carry over. Under Model A, without the tax-rate interaction, the index is −0.159 ($p = 0.34$) and the elderly-share interaction is 0.057 ($p = 0.007$).

Table B3: Charter Proximity Interactions

Variable	(1) × within 10 mi	(2) × within 15 mi	(3) × hosts campus charter
Hybrid fragmentation index	-1.7854^{**}	-1.0247	-1.3636^{**}
	(0.8111)	(0.8584)	(0.6028)
$HFI\times$ interaction flag	0.7819	-0.5163	-3.1189
	(0.9753)	(0.9965)	(3.9410)
p on the interaction	0.423	0.605	0.429
Full control set	Yes	Yes	Yes
Propositions	1,941	1,941	1,941

Notes: Columns (1) and (2) interact the index with an indicator for an open-enrollment charter campus within ten and fifteen miles respectively, computed from zip-code centroids using current charter locations. Column (3) interacts it with an indicator for the district hosting its own campus charter, which applies to thirteen districts contributing 73 propositions; in the corresponding logit that interaction is -52.14 ($p = 0.139$). With so few treated districts the specification has little power and we build nothing on it. $^{***}\,p < 0.01$, $^{**}\,p < 0.05$, $^{*}\,p < 0.10$. The interaction flags themselves are absorbed by the district fixed effects, since neither varies within a district.